ASTANA IT UNIVERSITY

«Approved»
by the Dean
__________ Syzdykova Z.A.
«___» of __________ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	**High Performance Computing (HPC)**
Degree cycle (level)/ major	Information and Communication Technologies
Year, term	3, 1
Number of credits	5
Language of delivery:	English
Prerequisites	Algorithms and Data structures Advanced programming
Postrequisites	Heterogeneous Parallelization
Lecturer(s)	Adai Shomanov, PHD, Associate Professor, **adai.shomanov@astanait.edu.kz**), Astana IT University, EXPO, C1 Block, 3rd floor, office C1.1.332.
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	This course offers a comprehensive exploration of how to optimize the performance of parallel computing systems, ranging from small multi-core processors to large-scale computing clusters. Key topics include parallel architecture design, efficient resource management, and strategies for optimizing memory access. Students will learn to analyze and design parallel algorithms, utilizing advanced programming techniques that can be applied across various hardware platforms. The course also emphasizes performance benchmarking and scalability analysis, enabling students to assess the effectiveness of their parallel implementations in diverse real-world applications.
2. Course Goal(s)	1. Understand Parallel Computing Architectures 2. Design and Implement Efficient Parallel Algorithms 3. Master Parallel Programming Tools and Techniques 4. Optimize Performance of Computational Applications 5. Apply High-Performance Computing to Real-World Problems 6. Evaluate and Benchmark Parallel Solutions
3. Course Objectives:	• Understand parallel computing concepts. • Analyze parallel system architectures. • Design and develop parallel algorithms. • Use parallel programming models and tools. • Optimize memory access and performance. • Assess scalability and efficiency of parallel solutions.
4. Skills & Competences	• Proficiency in parallel programming languages (OpenCL, CUDA, MPI). • Ability to design efficient parallel algorithms.

	• Competence in optimizing memory access and data flow in parallel systems. • Skills in analyzing and benchmarking performance of parallel applications. • Understanding of CPU and GPU architectures for high-performance computing. • Ability to solve complex computational problems using parallelism. • Knowledge of load balancing, synchronization, and task scheduling. • Expertise in debugging and optimizing parallel code. • Familiarity with performance profiling tools and techniques. • Capability to implement scalable solutions on multi-core and distributed systems.
5. Course Learning Outcomes:	By the end of this course students will be able to: • Explain the principles and architectures of modern parallel computing systems. • Design and implement parallel algorithms for multi-core CPUs and GPUs. • Utilize parallel programming frameworks such as OpenCL, CUDA, and MPI effectively. • Optimize parallel programs for performance, including memory management and data distribution. • Analyze the scalability and efficiency of parallel solutions in various computing environments. • Solve computationally intensive problems using advanced parallel processing techniques. • Evaluate and benchmark the performance of parallel applications across different hardware platforms.
6. Methods of Assessment	• Assignments • Midterm • Final Exam
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Basic Literature: 1. "Parallel Programming: Techniques and Applications Using Networked Workstations and Parallel Computers" by Barry Wilkinson and Michael Allen. 2. "CUDA by Example: An Introduction to General-Purpose GPU Programming" by Jason Sanders and Edward Kandrot. 3. "Programming Massively Parallel Processors: A Hands-on Approach" by David Kirk and Wen-mei Hwu. 4. "Introduction to Parallel Computing" by Ananth Grama, Anshul Gupta, George Karypis, and Vipin Kumar. Supplementary literature:

	1. "A Survey of General-Purpose Computation on Graphics Hardware" by John D. Owens et al. 2. "The Landscape of Parallel Computing Research: A View from Berkeley" by David Patterson et al. 3. Relevant papers from journals such as IEEE Transactions on Parallel and Distributed Systems or Journal of Parallel and Distributed Computing.
1. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
2. **Course policy**	**Course and University policies include:** **Attendance:** Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as "not graded". In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean's office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. **Class work**: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes. **Being late on class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning,

	and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. **Attestation I and II:** Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. **Home work / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. **Late submissions:** Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.* **Laptops and mobile devices** can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. **Online lessons** can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

	Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). **Contacting the Course instructor**: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.

* *Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Course Overview and Introduction to HPC	1	1	1	3	3
2	Fundamentals of Parallelism and Computing Systems	1	1	1	3	3
3	CPU Architectures for Parallel Computing	1	1	1	3	3
4	GPU Architectures and Heterogeneous Computing	1	1	1	3	3
5	Challenges in Designing Efficient Parallel Programs	1	1	1	3	3
6	Introduction to Parallel Programming Models	1	1	1	3	3
7	Memory Management and Data Distribution	1	1	1	3	3
8	API Functions and Parallel Programming Tools	1	1	1	3	3
9	SPMD Parallel Programming I	1	1	1	3	3
10	CUDA Programming Model	1	1	1	3	3

	Total hours: 90	**10**	**10**	**10**	**30**	**30**

3.2 List of Assignments for Student Independent Study

№	**Assignments (topics) for Independent study**	**Hours**	**Recommended literature and other sources (links)**	**Form of submission**
1	**2**	**3**	**4**	**5**
1	**Research the evolution of parallel computing systems and prepare a brief report on the impact of parallelism on modern computing.**	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
2	Analyze the differences between shared-memory and distributed-memory architectures and summarize key challenges in each.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
3	Implement a simple parallel algorithm on a multi-core CPU using threads or OpenMP and evaluate its performance.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
4	Investigate GPU architectures and write a comparative analysis of CPU vs. GPU processing in high-performance computing.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
5	Develop a small parallel program that addresses a real-world problem (e.g., matrix multiplication) and outline the challenges faced in synchronization and load balancing.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
6	Write a review of OpenCL and CUDA programming models, including their strengths and limitations, supported by examples from recent research.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
7	Prepare for the midterm by solving practical problems in parallel programming, focusing on optimization techniques for multi-core systems.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
8	Research data allocation strategies in high-performance computing and present a case study on an effective memory management technique in parallel applications.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
9	Explore key API functions in	9	Reading list, books,	Reading list,

	CUDA/OpenCL and implement a simple kernel to solve a mathematical problem (e.g., vector addition).		internet resources Electronic version	books, internet resources Electronic version
10	Develop a parallel application using the SPMD model, demonstrate the results, and discuss the potential for further optimization.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1st attestation	**Assignments**:** assignment 1 **Mid Term**	**40** **60**	100
2nd attestation	**Assignments**:** assignment 2 assignment 3 **End Term**	**40** 20 20 **60**	100
Final exam*	Quiz		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

** *The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.*

* *Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	

D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Shomanov Adai	Associate Professor	26/08/2024	

Director М. Сергазиев **Sergaziyev M.Zh.**